



Thakur Shree DPS College of Engineering & Management, Vasai (East)

First Year Engineering

Academic Year: 2026-2027

Assignment No: 01

Subject: BSC102/AP

Date: 07/10/2026

Max Marks: 10

Fundamental Constants:

- Planck's constant, $h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s} = 4.136 \times 10^{-15} \text{ eV} \cdot \text{s}$
 - Speed of light in vacuum, $c = 3.0 \times 10^8 \text{ m/s}$
 - Boltzmann constant, $k_B = 1.381 \times 10^{-23} \text{ J/K} = 8.617 \times 10^{-5} \text{ eV/K}$
 - Elementary charge, $e = 1.602 \times 10^{-19} \text{ C}$
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Problem 1: Transition Energy to Frequency and Wavelength

In a Ruby laser, chromium ions (Cr^{3+}) undergo radiative decay between an excited metastable state E_2 and the ground state E_1 . The energy difference between these two laser levels is $\Delta E = 1.79 \text{ eV}$.

- Calculate the frequency (ν) of the emitted radiation.
- Determine the wavelength (λ) of the emitted laser beam in nanometers (nm) and state which region of the electromagnetic spectrum it occupies.

Given:

$$\Delta E = E_2 - E_1 = 1.79 \text{ eV} = 1.79 \times 1.602 \times 10^{-19} \text{ J} \approx 2.868 \times 10^{-19} \text{ J}$$

(a) Frequency (ν): Using the Bohr-Einstein relation:

$$\nu = \frac{\Delta E}{h} = \frac{2.868 \times 10^{-19} \text{ J}}{6.626 \times 10^{-34} \text{ J} \cdot \text{s}} \approx 4.328 \times 10^{14} \text{ Hz} = 432.8 \text{ THz}$$

(b) Wavelength (λ): Using $c = \nu\lambda$:

$$\lambda = \frac{c}{\nu} = \frac{hc}{\Delta E} = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})(3.0 \times 10^8 \text{ m/s})}{2.868 \times 10^{-19} \text{ J}} \approx 6.931 \times 10^{-7} \text{ m} = 693.1 \text{ nm}$$

Spectral Region: The emitted radiation lies in the **red region** of the visible spectrum.

Problem 2: He-Ne Laser Transition

A Helium-Neon (He-Ne) gas laser produces a highly coherent beam through stimulated emission at a characteristic wavelength of $\lambda = 632.8$ nm.

- (a) Find the transition frequency (ν) of the emitted photons.
- (b) Calculate the energy difference ($E_2 - E_1$) between the upper and lower lasing levels in both Joules (J) and electron-volts (eV).

Given:

$$\lambda = 632.8 \text{ nm} = 632.8 \times 10^{-9} \text{ m}$$

(a) Frequency (ν):

$$\nu = \frac{c}{\lambda} = \frac{3.0 \times 10^8 \text{ m/s}}{632.8 \times 10^{-9} \text{ m}} \approx 4.741 \times 10^{14} \text{ Hz} = 474.1 \text{ THz}$$

(b) Energy Difference ($E_2 - E_1$):

$$\Delta E = h\nu = (6.626 \times 10^{-34} \text{ J} \cdot \text{s})(4.741 \times 10^{14} \text{ s}^{-1}) \approx 3.141 \times 10^{-19} \text{ J}$$

In electron-volts (eV):

$$\Delta E = \frac{3.141 \times 10^{-19} \text{ J}}{1.602 \times 10^{-19} \text{ J/eV}} \approx 1.961 \text{ eV}$$

Problem 3: Population Inversion

Consider a two-level atomic system with an energy separation corresponding to a transition wavelength $\lambda = 694.3$ nm (Ruby laser).

- (a) Calculate the ratio of the population of the upper level (N_2) to the lower level (N_1) at thermal equilibrium at room temperature ($T = 300$ K).
- (b) At what hypothetical temperature T would the thermal population of the upper level reach 10% of the ground state population ($N_2/N_1 = 0.1$)?

Given:

$$\lambda = 694.3 \text{ nm} = 6.943 \times 10^{-7} \text{ m}, \quad T = 300 \text{ K}$$

The transition energy is:

$$\Delta E = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})(3.0 \times 10^8 \text{ m/s})}{6.943 \times 10^{-7} \text{ m}} = 2.863 \times 10^{-19} \text{ J} \approx 1.787 \text{ eV}$$

The thermal energy at 300 K is:

$$k_B T = (1.381 \times 10^{-23} \text{ J/K})(300 \text{ K}) = 4.143 \times 10^{-21} \text{ J} \approx 0.02586 \text{ eV}$$

(a) Ratio N_2/N_1 at 300 K:

$$\frac{\Delta E}{k_B T} = \frac{2.863 \times 10^{-19} \text{ J}}{4.143 \times 10^{-21} \text{ J}} \approx 69.10$$

$$\frac{N_2}{N_1} = \exp\left(-\frac{\Delta E}{k_B T}\right) = e^{-69.10} \approx 9.77 \times 10^{-31} \approx 10^{-30}$$

Physical significance: At room temperature, the upper excited state is virtually empty.

(b) Temperature for $N_2/N_1 = 0.1$:

$$\ln\left(\frac{N_2}{N_1}\right) = -\frac{\Delta E}{k_B T} \implies T = \frac{\Delta E}{k_B \ln(N_1/N_2)}$$

Since $N_1/N_2 = 10$:

$$T = \frac{2.863 \times 10^{-19} \text{ J}}{(1.381 \times 10^{-23} \text{ J/K}) \cdot \ln(10)} = \frac{2.863 \times 10^{-19}}{1.381 \times 10^{-23} \times 2.3026} \approx 8997 \text{ K} \approx 9000 \text{ K}$$

Problem 4: Determination of Excited State Energy Level

In a gas laser medium at thermodynamic equilibrium at a temperature of $T = 500 \text{ K}$, the lower energy level is chosen as the reference state ($E_1 = 0 \text{ eV}$) with population N_1 . The population of an excited level E_2 is measured to be $N_2 = 1.25 \times 10^{-4} N_1$.

(a) Using the Boltzmann distribution $N(E) = N_0 \exp(-E/k_B T)$, calculate the energy of the upper level E_2 in electron-volts (eV).

Given:

$$T = 500 \text{ K}, \quad \frac{N_2}{N_1} = 1.25 \times 10^{-4}, \quad k_B = 8.617 \times 10^{-5} \text{ eV/K}$$

(a) Energy of Upper Level with $E_1 = 0 \text{ eV}$:

From the Boltzmann ratio between two non-degenerate states:

$$\frac{N_2}{N_1} = \exp\left(-\frac{E_2 - E_1}{k_B T}\right)$$

Taking the natural logarithm on both sides:

$$\ln\left(\frac{N_2}{N_1}\right) = -\frac{E_2 - E_1}{k_B T} \implies E_2 - E_1 = k_B T \ln\left(\frac{N_1}{N_2}\right)$$

Evaluating the terms:

$$k_B T = (8.617 \times 10^{-5} \text{ eV/K})(500 \text{ K}) = 0.043085 \text{ eV}$$

$$\ln\left(\frac{N_1}{N_2}\right) = \ln\left(\frac{1}{1.25 \times 10^{-4}}\right) = \ln(8000) \approx 8.9872$$

Therefore:

$$E_2 - E_1 = (0.043085 \text{ eV}) \times 8.9872 \approx 0.3872 \text{ eV}$$

Since $E_1 = 0 \text{ eV}$:

$$E_2 = 0.3872 \text{ eV} \approx 0.39 \text{ eV}$$